Relational Gravitational Dynamics: a single-amplitude radial term, tests of $g^\dagger(z)$, and the cosmological limits of the kernel

Unifying disk galaxies, galaxy groups and galaxy clusters under a single effective law — second paper

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Abstract

The effective gravitational kernel of the previous version (v4) describes the ratio of observed to baryonic acceleration in bound systems with $K_{\rm eff} = 1 + \lambda e^{-x}/(1-e^{-x})^n$, $x = g_{\rm bar}/g^\dagger(z)$, $g^\dagger = cH(z)/2\pi$ and $n(\xi) = \frac{1}{2} + A\xi$. Its radial extension, an exponent modulation $w^{-\alpha(\xi)}$ with two extra parameters calibrated on a subset of the same data, was left as a semi-predictive result. Here the radial dependence is replaced by a scale shift $s = [\Delta(r)/\Delta_{500}]^{\,p\,\Theta(\xi)}$, $\Delta/\Delta_{500} = \bar\rho(<r)/500\rho_c$, with a single radial amplitude p and a logistic switch Θ in the compactness ξ (threshold ξ_c, width m) that turns the term on in groups and clusters and off in galaxies. On the 39 X-COP radial points of v4 the rms falls from 0.037 to 0.021 dex at $p = 0.5$ (0.016 at the fitted $p = 0.45$); the cluster-bootstrap 95% interval of the paired difference is 0.007–0.024 dex, and the comparison is in-sample. With $p = 0.5$ frozen, other samples were predicted without refitting. In ten Vikhlinin et al. (2006) clusters the ratio $f_{\rm bar}(r_{2500})/f_{\rm bar}(r_{500})$ is predicted with $\ln(\text{obs/pred}) = -0.10 \pm 0.06$ (v4 law -0.61 ± 0.21; no radial term -1.29 ± 0.07), and in nine X-COP clusters $f_{\rm bar}(r_{200})/f_{\rm bar}(r_{500})$ with $+0.03 \pm 0.04$ (v4 law $+0.02 \pm 0.06$; none $+0.87 \pm 0.06$). The best-fit p nevertheless differs between samples (0.45–0.60) by more than the statistical errors. A hierarchical model of the per-system amplitudes gives $p_0 = 0.53 \pm 0.01$ with an intrinsic scatter $\tau = 0.05$ among clusters (about 13% in the predicted baryon fraction) and 0.17 among groups; the mass dependence of p seen in groups is absorbed by the ξ switch, and no residual dependence on mass, redshift or instrument is significant individually. A hydrostatic bias of 0–30% moves p by less than 0.1. The galaxy–group switch (supported at $\Delta(-2\ln L) \simeq 28$ by the hierarchical model) is loosely constrained: widths $m \geq 3$ and thresholds $0.09 \lesssim \xi_c \lesssim 0.16$ fit comparably, and a smooth power law $p(\xi)$ without threshold is disfavoured by $\Delta(-2\ln L) \simeq 50$. The evolution $g^\dagger \propto H(z)$ is preferred by measured gas fractions (a frozen $g^\dagger$ is rejected by $\Delta\chi^2 = 57$ and 48), whereas median KMOS^{3D} galaxies at $z \simeq 0.9$–2.3 favour a weaker evolution, a difference within the $\sim 10\%$ systematics. We document unresolved tensions: the baryon fraction at r_{2500} rises with mass more steeply than observed, the absolute normalisation differs by $\sim 20\%$ between the Chandra and XMM samples, and the high-redshift galaxy trend. Applied to linear cosmological perturbations the kernel boosts gravity by factors 10^2–10^3; a density-based damper restores ΛCDM growth only by postulating the plateau $K_c = \Omega_m/\Omega_b$. The program is an explicitly phenomenological map of bound systems, comparable to ΛCDM in clusters, parsimonious but poorer in fit in galaxies, and silent on the CMB.

Keywords: dark matter alternatives · modified gravity · galaxy clusters · radial acceleration relation · baryon fraction · Tully–Fisher relation · cosmological evolution of $g^\dagger$

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1 Introduction

The kernel proposed in the first paper of this program (v4, hereafter Paper I) maps the baryonic acceleration $g_{\rm bar}$ of a bound system to its observed acceleration, $g_{\rm obs} = K_{\rm eff}\, g_{\rm bar}$, using a transition scale $g^\dagger(z) = cH(z)/2\pi$ and an exponent n that depends on the equilibrium geometry through the compactness $\xi = g_{500}/g^\dagger(z)$. Paper I tested it on SPARC disk galaxies, 23 groups, 13 X-COP clusters and 23 CLASH clusters and listed its own limitations. Four of them are the subject of this work.

1. *The radial law was semi-predictive.* The factor $w^{-\alpha(\xi)}$, with $\alpha(\xi) = 0.355(1 - e^{-\xi/0.052})$, used two parameters derived earlier from a subset of eight X-COP clusters, and the 39-point test of Paper I partially overlaps that calibration. Paper I lists validation on resolved profiles, with a falsification criterion of 0.10 dex, as an open item.
2. *The evidence for $g^\dagger \propto H(z)$ rested on CLASH gas fractions estimated from a scaling relation.* Paper I itself discusses whether part of the residual correlation is algebraic.
3. *Groups were 23 and no variable switched the radial term between galaxies and clusters.*
4. *No test at high redshift and no treatment of cosmological scales* were given; the extension was declared outside the scope.

We (i) replace the radial law by a scale shift with a single amplitude and compare it with the v4 law on the same data and, with the amplitude frozen, on other samples (Sect. 4); (ii) test $g^\dagger(z)$ with measured gas fractions and with $z \simeq 1$–2 galaxies (Sect. 5); (iii) introduce a switch in ξ connecting galaxies to groups and clusters, test whether it must be a sharp step, and derive its redshift dependence (Sect. 6); (iv) compare with ΛCDM (Sect. 7); (v) document the tensions (Sect. 8); and (vi) examine what happens when the kernel is applied to cosmological perturbations and how a density-based damper behaves (Sect. 9). Section 10 states what the program does and does not claim, and lists falsifiable tests.

2 The kernel and its radial extension

2.1 Definition

For a bound, spherically or axially symmetric system in equilibrium,

$$K_{\rm eff} = 1 + \frac{\lambda\, e^{-x/s}}{\left(1 - e^{-x/s}\right)^n}, \qquad x = \frac{g_{\rm bar}}{g^\dagger(z)}, \qquad g^\dagger(z) = \frac{cH(z)}{2\pi}, \tag{1}$$

with $\lambda = 0.999$, $H_0 = 70\ {\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.3$, and the exponent law of Paper I,

$$n(\xi) = \tfrac{1}{2} + A\,\xi, \qquad A = 0.4068, \qquad \xi = \frac{g_{500}}{g^\dagger(z)} = \frac{2\pi\, GM_{500}}{c\,H(z)\,R_{500}^2}. \tag{2}$$

For $s = 1$ and $\xi \to 0$ this is the disk-galaxy kernel ($n = 1/2$) with the baryonic Tully–Fisher relation $v_f^4 = \lambda^2 g^\dagger G M_b$ as a structural consequence. The compactness ξ scales as $(MH)^{1/3}$.

2.2 Radial term

Paper I modulated the exponent, $n \to n(\xi)\, w^{-\alpha(\xi)}$ with $w = r/R_{500}$. Here the exponent is kept at its $w = 1$ value and the radial dependence enters through the scale s,

$$s = \left[\frac{\Delta(r)}{\Delta_{500}}\right]^{p\,\Theta(\xi)}, \qquad \frac{\Delta(r)}{\Delta_{500}} = \frac{\bar\rho(<r)}{500\,\rho_c(z)} = \frac{M(<r)}{M_{500}}\left(\frac{R_{500}}{r}\right)^3. \tag{3}$$

By definition $\Delta/\Delta_{500} = 1$ at r_{500}, 5 at r_{2500} and 0.4 at r_{200}, so p is the only radial amplitude and the $w = 1$ calibration of Paper I is untouched. $\Theta(\xi) = [1 + (\xi_c/\xi)^m]^{-1}$ is a logistic switch in $\ln \xi$. Sect. 6 shows that the data constrain neither its sharpness nor its location tightly ($m \geq 3$, $0.09 \lesssim \xi_c \lesssim 0.16$); the fiducial values are $m = 4$, $\xi_c = 0.11$. For clusters ($\xi \geq 0.34$) $\Theta \geq 0.98$ for every acceptable choice, so the cluster results below do not depend on it; systems with $\xi \ll \xi_c$ have $s = 1$.

2.3 Self-consistent baryon fraction

In a system with total enclosed mass $M(< r)$ and baryonic fraction $f = M_{\rm bar}/M$, $g_{\rm obs} = GM/r^2$ and $x = f\, g_{\rm obs}/g^\dagger$; the identity $K = g_{\rm obs}/g_{\rm bar} = 1/f$ gives the equation $f\, K(f\, g_{\rm obs}/g^\dagger) = 1$, solved for f. The prediction is a baryon fraction, which is compared with the observed gas plus stellar fraction. Because the observable depends on the measured total mass, hydrostatic masses are interpreted as the effective dynamical mass $g_{\rm obs} r^2/G$; they are unbiased with respect to the gravity law if the gas is in equilibrium in the potential of Eq. (1) (Sect. 8).

3 Data and method

Table 1 lists the samples. Tables for Allen, Mantz, Vikhlinin, Lovisari, Sun, Ettori and Umetsu were transcribed from the published papers (Appendix A) and cross-checked where redundancy allows (e.g. Mantz M_{2500} against the value implied by r_{2500} and the critical density, median ratio 0.999; Vikhlinin M_{500} against $500\rho_c(4\pi/3)r_{500}^3$, 0.96–1.02).

Sample	N	Observable	Use
SPARC (Lelli et al. 2016), Table 2	175 gal. (2803 pts after cuts)	rotation curves	galaxies, $s = 1$
Lovisari+15, Sun+09	43 groups	M, $M_{\rm gas}$ at r_{500}, r_{2500}	transition regime
X-COP (Ettori+19, Eckert+19, Ghirardini+19)	13 / 12 clusters	$M(< r)$ at 0.5, 1, 1.5 Mpc / $f_{\rm gas}(r_{500}, r_{200})$	radial term
Vikhlinin+06	12 (10 with both radii)	M_{2500}, cumulative $f_{\rm gas}$ at r_{2500}, r_{500}	radial term at r_{2500}
Allen+08	42	cumulative $f_{\rm gas}(r_{2500})$, $z \leq 1.1$	$g^\dagger(z)$
Mantz+14	40	shell $f_{\rm gas}$ at 0.8–1.2 r_{2500}, z = 0.08–1.06	$g^\dagger(z)$
CLASH (Umetsu+16)	12 overlapping	lensing M_{2500}, M_{500}	HSE vs. lensing
KMOS$^{\rm 3D}$ (Übler+17)	135 gal. (3 bins)	$M_{\rm bar}$, R_e, $v_{\rm circ,max}$ at $z \simeq 0.9, 1.5, 2.3$	high-z galaxies

Table 1: Samples. For SPARC, points with $V > 0$, $R > 0$ and $\sigma_V/V < 0.1$ are used, with $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$ for all models; the comparison with halos uses the 141 galaxies with ≥ 5 points (2736 points).

Statistics. Residuals are in $\log_{10}$ (dex) unless stated as ln; model comparison uses $-2\ln L = \sum N \ln(\mathrm{rms}^2)$ with AIC $= \sum N \ln(\mathrm{rms}^2) + 2k$. Points at different radii of one cluster, and cumulative profiles, are strongly correlated, so we also give results with one independent unit per cluster ($N_{\rm eff} = 13$ for X-COP; 141 galaxies and 43 groups in joint fits), and we count k in two ways: only the parameters added in the comparison, and in addition p, ξ_c, m and A (Sect. 7). Intervals on rms values, paired differences and best-fit p are cluster bootstraps (2000–4000 resamplings of the clusters). When the observable measures only the redshift or radial

shape (Mantz shells, Allen cumulative fractions in the α scan) the normalisation is a free scalar. Cluster stellar masses follow Kravtsov et al. (2018) (the relation of Gonzalez et al. 2013 was used only as an alternative in cross-checks), $\log M_{*,\rm tot} = 12.71 + 0.59(\log M_{500} - 14.5)$ (0.11 dex scatter, $M_{500} = 4.6 \times 10^{13}$–$1.6 \times 10^{15}\,M_\odot$), with a fraction φ of it inside r_{2500} ($\varphi = 0.7$, 1.0 inside r_{500}). A stellar fraction of 9% of the gas, used in the first draft for groups, is kept as a sensitivity case. φ has not been measured and is varied (0.5–1).

3.1 Provenance of the parameters and independence of the tests

Table 2 states what was fixed on which data. Two consequences matter. First, A was calibrated in Paper I on 23 groups, the 13 X-COP clusters and 23 CLASH clusters at r_{500}, so the prediction *at* r_{500} is not independent for those samples; only the radial behaviour is a prediction. Second, the samples overlap (A1795 and A2029 are in Vikhlinin and X-COP, 12 clusters are in CLASH and Allen, Allen and Mantz share several), so tests on different samples are not statistically independent.

Quantity	Value	Determined on	Consequence for the tests here
λ, $g^\dagger(0)$	0.999, $cH_0/2\pi$	SPARC global fit (Paper I)	independent of the cluster and group samples
α in $g^\dagger \propto E^\alpha$	1 (assumed)	not fitted	tested in Sect. 5
A	0.4068	59 systems of Paper I (23 groups, 13 X-COP, 23 CLASH) at r_{500}	r_{500} predictions for these samples not independent
p	0.45 (fit), 0.5 (adopted)	X-COP, 39 points	Vikhlinin, Ettori r_{200}, groups and Allen are held out with p frozen (Sect. 4.3)
ξ_c, m	0.09–0.16, ≥ 3	SPARC (ξ imputed from a halo model), groups, X-COP	loosely constrained (Sect. 6)
φ	0.7 (0.5–1)	external relation; fraction inside r_{2500} not measured	systematic for all r_{2500} results
H_0, Ω_m	70, 0.3	fixed	source cosmologies not propagated

Table 2: Provenance of the quantities that are not refitted in each test.

4 The radial term: comparison with the v4 law

4.1 Same data as Paper I

We reproduce the v4 test on the 39 X-COP points (13 clusters at $r = 0.5, 1.0, 1.5$ Mpc, $w = 0.35$–1.66, $f_{\rm bar} = 1.09\, f_{\rm gas}(M_{500})$ from the Sun et al. scaling relation as in Paper I). The law $n(\xi)w^{-\alpha(\xi)}$ gives 0.037 dex (Paper I: 0.040; the small difference may come from details of the input masses and baryon fraction that we could not reproduce exactly), fixing n at $w = 1$ gives 0.126, Newton alone 0.878, MOND 0.257 (Table 3).

Uncertainties. Resampling the 13 clusters (4000 draws) gives rms 0.126 [0.111, 0.140] without radial term, 0.037 [0.026, 0.047] for the v4 law, 0.021 [0.014, 0.027] for $p = 0.5$ and 0.016 [0.012, 0.019] for the fitted $p = 0.45$ [0.40, 0.50] (95% intervals). The paired difference rms(v4)$-$rms($p = 0.5$) is $+0.016$ [0.007, 0.024] dex and is positive in every draw. Counting one independent unit per cluster ($N_{\rm eff} = 13$) the AIC is -54, -82, -101 and -105 for no radial term, v4, $p = 0.5$ and fitted p (against -162, -253, -301, -321 with $N = 39$): the ordering is unchanged and the margin of the new term over v4 is 19–23. Charging p (and ξ_c, m, A as in Sect. 7) lowers these margins by at most 8.

Model	extra par.	rms [dex]	0.5 Mpc	1.0 Mpc	1.5 Mpc	AIC
$n(\xi)$ fixed at $w = 1$	0	0.126	0.197	0.062	0.071	−162
v4: $n(\xi)\, w^{-\alpha(\xi)}$	2	0.037	0.062	0.013	0.011	−253
This work, $p = 0.5$ fixed	0	0.021	0.035	0.006	0.008	−301
This work, $p = 0.45$ fitted	1	0.016	0.024	0.007	0.012	−321
NFW + Duffy $c(M)$, M_{200} free	13	0.039	—	—	—	−227

Table 3: X-COP, 39 points. Δ/Δ_{500} takes median values 5.2, 1.5 and 0.67 at the three radii. A leave-one-cluster-out fit gives $p = 0.45$–0.46 for every omitted cluster and a predictive rms of 0.017 dex. The v4 exponent modulation and the new scale shift are both calibrated on this sample, so this comparison is in-sample. AIC here counts only the parameters added; see the text for the charge of p, ξ_c, m, A and for $N_{\rm eff} = 13$.

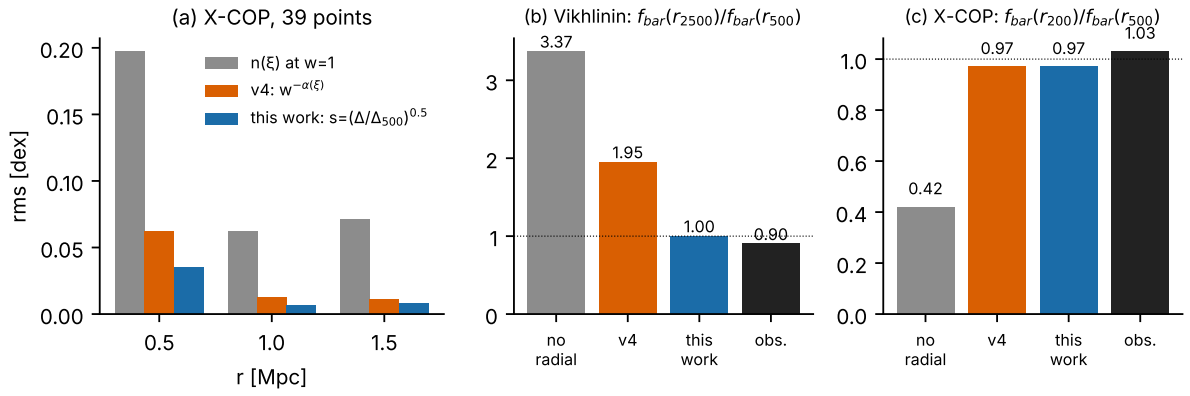


Figure 1: (a) rms residual by physical radius in the X-COP sample. (b) Predicted and observed $f_{\rm bar}(r_{2500})/f_{\rm bar}(r_{500})$ in ten Vikhlinin et al. clusters. (c) Same ratio at r_{200}/r_{500} in nine X-COP clusters with measured $f_{\rm gas}$ (Ettori et al. 2019). "no radial" denotes $n(\xi)$ at $w = 1$ with $s = 1$.

4.2 Independent data

Vikhlinin et al. (2006), r_{2500}. For the ten clusters with cumulative $f_{\rm gas}$ at both radii ($\xi = 0.23$–0.59, $M_{500} = 7.4 \times 10^{13}$–$1.1 \times 10^{15}\, M_\odot$, $w = r_{2500}/r_{500} = 0.39$–$0.48$) the control at r_{500}, where the radial term plays no role, gives obs/pred = 0.95 (rms 0.13 in ln). At r_{2500} without a radial term obs/pred is 0.26 ($\rm rms_{ln}$ 1.35, predicted $f_{\rm bar}(r_{2500})/f_{\rm bar}(r_{500}) = 3.37$); with the v4 law 0.45 ($\rm rms_{ln}$ 0.90, ratio 1.95, and one of the ten clusters has no solution because $n > 1$ inside $w \simeq 0.45$); with the scale shift 0.84 ($\rm rms_{ln}$ 0.23, ratio 1.00). The observed ratio is 0.90. The v4 law, which scales the exponent as $w^{-\alpha}$, drives n beyond 1 at $w < 1$ for the highest ξ, where the kernel no longer has a finite baryon fraction; the scale shift cannot do this because it multiplies x rather than the exponent. At the criterion proposed in Paper I (0.10 dex), the v4 law with its published $\alpha(\xi)$ fails on this sample (we did not re-optimise α).

X-COP out to r_{200}. Ettori et al. (2019, Table 2) give $f_{\rm gas}$ and $M_{\rm HSE}$ at r_{500} and r_{200} for 12 clusters. After adding stars (Kravtsov scaling, $\varphi = 1$) the ratio $f_{\rm bar}(r_{200})/f_{\rm bar}(r_{500})$ is observed to be 1.03 (median); the scale shift with $p = 0.5$ predicts 0.98 (all 12) or 0.97 (the nine clusters left after excluding the mergers A2319, A3266 and ZW1215), against 0.97 for the v4 law (rms of ln[obs/pred ratio] 0.157 against 0.120 for the new term) and 0.42 without a radial term. The mass trend of this ratio is −0.12 per dex observed and −0.11 predicted. The mass-dependent exponent $p(\xi) \propto \xi^{0.37}$ (Sect. 8) predicts +0.11 and is disfavoured. At r_{500} itself, the X-COP sample has obs/pred = 1.17 (rms 0.19 in ln) while the Vikhlinin sample gives 0.95 (Sect. 8).

Gain of the radial term. Without the radial term the kernel overestimates the baryon fraction at r_{2500} by a factor 3–4 (obs/pred 0.27, $\mathrm{rms}_{\ln}$ 1.27 over 12 clusters) and underestimates it at r_{200} by a factor 2.4 (ratio 0.42 against 1.03). With $p = 0.5$ the rms falls to 0.225 at r_{2500} and 0.12 at r_{200}. In galaxies $\xi < \xi_c$, so $s = 1$ and the term is inactive: the gain belongs to groups and clusters. The test shows that the term is necessary and that $[\Delta/\Delta_{500}]^p$ describes it with one parameter; it does not show that this is the only possible form, and $p \simeq 0.5$ was calibrated on samples that partially overlap (A1795 and A2029 are in both Vikhlinin and X-COP; see Sect. 4.3 for the held-out comparison).

4.3 Frozen-p predictions and the dispersion of p

With p fixed on X-COP (0.45, or its rounded value 0.5) no parameter was refitted for the four samples of Table 4. The residual is $\ln(f_{\rm bar}^{\rm obs}/f_{\rm bar}^{\rm pred})$ at the stated radius with the self-consistent $f_{\rm bar}$ of Sect. 2, mean and rms over clusters; the v4 law is used with its published $\alpha(\xi)$ and was not re-optimised.

Sample (radius, N)	no radial term	v4 law	new, $p = 0.45$	new, $p = 0.50$	best-fit p [95%]
Vikhlinin (r_{2500}, 10)	$-1.33/1.35$	$-0.66/0.90^a$	$-0.28/0.33$	$-0.15/0.23$	0.55 [0.50, 0.60]
X-COP Ettori (r_{200}, 9)	$+1.03/1.04$	$+0.18/0.22$	$+0.27/0.29$	$+0.19/0.21$	0.60 [0.60, 0.65]
Groups (r_{2500}, 43)	$-0.82/0.92$	$+0.66/1.45$	$0.00/0.40$	$+0.09/0.41$	0.45 [0.39, 0.50]^b
Allen (r_{2500}, 42)	$-1.35/1.37$	$-0.83/0.91^c$	$-0.22/0.30$	$-0.06/0.20$	0.55 [0.50, 0.55]
X-COP 39 pts (fit sample)	—	—	—	—	0.45 [0.40, 0.50]

Table 4: Mean/rms of $\ln(f_{\rm bar}^{\rm obs}/f_{\rm bar}^{\rm pred})$ with p frozen. These include each sample's zero point (see the ratio test in the text). ^{a}One cluster without solution ($n > 1$). ^{b}With Kravtsov stars, $\varphi = 0.7$; with 9% of the gas the groups need $p = 0.80$ [0.76, 0.85]. ^{c}Three clusters without solution.

Zero-point-free ratio test. Dividing out the control at r_{500} of each sample (where p plays no role: -0.05 ± 0.04 for Vikhlinin and $+0.16 \pm 0.02$ for Ettori) gives the mean of ln of the ratio between the observed and the predicted $f_{\rm bar}(r)/f_{\rm bar}(r_{500})$. At r_{2500} in ten Vikhlinin clusters: -1.29 ± 0.07 (no radial term), -0.61 ± 0.21 (v4), -0.24 ± 0.06 ($p = 0.45$), -0.10 ± 0.06 ($p = 0.5$). At r_{200} in nine X-COP clusters: $+0.87 \pm 0.06$, $+0.02 \pm 0.06$, $+0.10 \pm 0.04$, $+0.03 \pm 0.04$ (errors are standard errors over clusters). The new term is clearly better than v4 inside r_{500} (Vikhlinin, groups, Allen, where v4 loses solutions or diverges) and equivalent to it between r_{500} and r_{200}, where v4 is calibrated.

Dispersion of p. The best-fit amplitudes range from 0.45 to 0.60 and their bootstrap intervals do not overlap (X-COP 0.40–0.50 against Ettori 0.60–0.65), so the scatter is systematic, not statistical: the five values have mean 0.52 and standard deviation 0.07. Near r_{2500} and r_{200} a change of 0.05 in p shifts the predicted $f_{\rm bar}$ by about 0.13 in ln (Vikhlinin: $-0.28 \to -0.15$ from $p = 0.45$ to 0.50), so the amplitude is known to $\sim \pm 20\%$ in the predicted baryon fraction when samples are compared. It is not a constant at the level of the current data; the hierarchical model of Sect. 4.4 separates an intrinsic scatter of 0.05 among clusters from a larger one among groups, and sources of the sample-to-sample differences are the hydrostatic bias (Sect. 8), the instrument zero points and the unmeasured stellar mass.

What is, and is not, held out. p was fixed on X-COP before the other samples were evaluated, but the tests are not blind: variants of the model (a mass-dependent $p(\xi)$, Sect. 8) were evaluated on the same data, the Kravtsov stellar treatment of the groups was adopted after

the 9% case was inspected, and the samples overlap. A blind test requires data not used here. The frozen prescription (parameters and code) is archived with this paper (Appendix A) so that it can be applied to new samples before they are examined.

4.4 A hierarchical model for the amplitude

To separate intrinsic scatter in p from measurement error and from covariates we solved, for every system with a measured gas fraction, the exponent p_i that reproduces its observed $f_{\rm bar}$ at the test radius (Vikhlinin r_{2500}, Ettori r_{200} with all 12 clusters, Allen r_{2500}, 43 groups at r_{2500}; 102 systems, 92 distinct clusters), and propagated an error σ_i from a 10% error on $f_{\rm bar}$ (Allen: the published error, median 13%) and a 10% error on the mass. The model is

$$p_i = \mu + \mathbf{x}_i \cdot \boldsymbol{\beta} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \tau^2 + \sigma_i^2),$$

with covariates $\mathbf{x}_i = (\log M_{500},\, z,\, \mathrm{XMM},\, \log \Delta/\Delta_{500})$ (XMM= 1 for X-COP and the Lovisari groups, 0 for Chandra samples), fitted by maximum likelihood; intervals on τ are profile likelihoods and on β cluster bootstraps over the 92 distinct clusters. In a second version μ is replaced by $p_0\Theta(\xi_i)$, the switch of Sect. 6, in which case p_i is the effective exponent $p\Theta$.

Model	N	μ or p_0	τ [95%]	homogeneity χ^2
constant p, all systems	102	0.497 ± 0.015	0.142 [0.122, 0.168]	659 / 101
constant p, clusters	59	0.526 ± 0.009	0.046 [0.028, 0.068]	129 / 58
constant p, groups	43	0.448 ± 0.032	0.200 [0.163, 0.255]	514 / 42
switch ($m = 8$, $\xi_c = 0.157$), all	102	0.532 ± 0.014	0.120	
switch, clusters	59	0.527 ± 0.009	0.046	
switch, groups	43	0.533 ± 0.032	0.174	

Table 5: Hierarchical model of the per-system amplitude p_i. τ is the intrinsic scatter in p. The last column is χ^2/d.o.f. of the fixed mean with measurement errors only.

Covariate	all (constant p)	clusters	groups	all, after switch
$\log M_{500}$ (per dex)	$+0.099 \pm 0.026$	$+0.018 \pm 0.041$	$+0.34 \pm 0.11$	-0.002 ± 0.023
z	$+0.05 \pm 0.08$	-0.104 ± 0.046	$+3.0 \pm 1.3$	-0.089 ± 0.074
XMM	-0.078 ± 0.032	$+0.045 \pm 0.027$	-0.166 ± 0.059	-0.021 ± 0.028
$\log \Delta/\Delta_{500}$	-0.091 ± 0.044	-0.041 ± 0.024	—	-0.045 ± 0.037

Table 6: Coefficients β of each covariate added alone. For clusters only, XMM and $\log \Delta/\Delta_{500}$ are confounded with the sample (Ettori is the only XMM cluster sample and the only one at r_{200}).

Results. (i) The amplitude is heterogeneous: with measurement errors alone the homogeneity χ^2 is 659 for 101 degrees of freedom (still 337 if the errors are inflated by 50%), so a pure constant is excluded and the intrinsic scatter is $\tau = 0.14$ [0.12, 0.17], i.e. 28% of p. (ii) The scatter is not uniform: among clusters it is $\tau = 0.046$ [0.028, 0.068] around $\mu = 0.526 \pm 0.009$, which corresponds to $\sim 13\%$ in the predicted baryon fraction (using $\partial \ln f_{\rm bar}/\partial p \simeq 2.7$ near r_{2500} and r_{200}); among groups it is 0.17–0.20 ($\sim 50\%$), where the stellar mass is not measured and the sample mixes two instruments. (iii) The correlation of p_i with mass ($+0.099 \pm 0.026$ per dex, $\Delta(-2\ln L) = 13.9$) comes from the groups ($+0.34 \pm 0.11$) and is absent among clusters ($+0.018 \pm 0.041$); once the ξ switch is included it vanishes (-0.002 ± 0.023). The switch ($m = 8$, $\xi_c = 0.157$ [0.138, 0.171] by bootstrap, $p_0 = 0.534$ [0.511, 0.559]) improves on a constant by $\Delta(-2\ln L) = 27.6$ for two parameters (bootstrap median 31, lower 2.5% quantile 9.9), an independent confirmation of the threshold of Sect. 6 that does not use galaxies; its

location is higher than in the joint fit with SPARC, which is consistent with the flat likelihood found there. (iv) Redshift, instrument and radius do not carry significant residual dependence individually after the switch (-0.089 ± 0.074, -0.021 ± 0.028, -0.045 ± 0.037); jointly they improve the likelihood by 10.7 for four degrees of freedom ($p \simeq 0.03$). Before the switch, the four-covariate fit gives bootstrap intervals for z (-0.13 [-0.26, -0.05]), XMM (-0.14 [-0.25, -0.03]) and $\log \Delta/\Delta_{500}$ (-0.15 [-0.29, -0.01]) that exclude zero, but these are collinear with sample membership and are not robust to the switch.

Correlation of p with redshift. Across the 59 clusters (Vikhlinin, Ettori, Allen; $z = 0.02$–1.06) the per-system amplitude anticorrelates with z: Spearman $\rho = -0.38$ ($p = 0.003$; bootstrap 95% interval $[-0.61, -0.11]$), hierarchical slope $\beta_z = -0.104 \pm 0.046$ per unit z (cluster bootstrap $[-0.20, -0.01]$) at $\alpha = 1$, and the slope does not change when $\log M_{500}$ is added. The correlation is between samples, not within one: the low-redshift clusters (Vikhlinin, Ettori; $\mu = 0.56 \pm 0.02$ for $z < 0.15$) have a higher p than the Allen clusters (0.51), and inside the Allen sample alone, the only one spanning $z = 0.06$–1.06 with one instrument and one radius, the correlation is absent ($\rho = -0.19$, $p = 0.26$; $\rho = +0.12$ for $z > 0.3$) and the slope is -0.037 ± 0.036 ($\tau \simeq 0$), i.e. $|\Delta p| < 0.07$ (2σ) between $z = 0$ and 1. The cluster-only slope depends on α: -0.08 to -0.13 for $0 \leq \alpha \leq 1.5$ and -0.19 ± 0.05 at $\alpha = 2$ (Allen alone: -0.02 to -0.05 for $\alpha \leq 1.5$, -0.11 ± 0.04 at $\alpha = 2$), vanishing only for no value of α in the cluster-only fit. We therefore find no evidence for an evolution of p with redshift within one sample; the cross-sample trend is equally an offset between instruments, radii and mass-estimation methods (Sect. 4.3), and a joint (α, β_z) determination needs a single homogeneous sample over a wider redshift range.

Robustness and limits. τ stays within 0.12–0.15 when the errors are halved or multiplied by 1.5, when the three X-COP mergers are dropped (0.137), when only one measurement per cluster is kept (0.150, 92 systems) or when $g^\dagger \propto E^{0.5}$ or $E^{1.5}$ is used (0.147, 0.141). The coefficient of z is degenerate with the exponent α of $g^\dagger(z)$: for clusters only it is -0.104 ± 0.046 at $\alpha = 1$ (2.3σ) and it changes with α, so a trend of p with redshift cannot be separated from a mis-specified evolution of $g^\dagger$ with these data. The Gaussian form of ε, the assumed σ_f and σ_M, the single radius per system and the sample overlap (handled only by the cluster bootstrap) limit the precision of τ; a per-system p_i also absorbs any error in the baryon fraction.

5 Evolution of $g^\dagger$

We parametrise $g^\dagger(z) = g^\dagger(0)\, E(z)^\alpha$, with $\alpha = 1$ the prediction $g^\dagger = cH/2\pi$ and $\alpha = 0$ a frozen scale. In clusters α enters both x and ξ, so a change of α moves the exponent as well as the transition.

5.1 Clusters: measured gas fractions

Allen et al. (2008, 42 clusters, cumulative $f_{\rm gas}(r_{2500})$) and Mantz et al. (2014, 40 clusters, shell $f_{\rm gas}$) are fitted with $p = 0.5$, $c_{500} = 3$ (to convert M_{2500} to M_{500} for ξ), a stellar addition of 9% and a free normalisation. Table 7 shows $\chi^2(\alpha)$.

A frozen $g^\dagger$ ($\alpha = 0$) is rejected with $\Delta\chi^2 = 57$ (Allen) and 48 (Mantz) relative to $\alpha = 1$, and $\alpha = 1$ is 5.4 and 1.5 above the respective minima. With $p = 0.46$ the pattern is the same ($\chi^2 = 82.8$ and 134.1 at $\alpha = 1$). In the earlier version of this test, in which only the kernel argument was frozen and ξ still followed $H(z)$, the Allen χ^2 was 267 against 69; the numbers in Table 7 are the consistent version.

α	0	0.25	0.50	0.75	1.00	1.25	1.50
Allen χ^2 (41 d.o.f.)	126.1	101.6	85.9	75.8	**69.4**	65.7	64.0
Mantz χ^2 (39 d.o.f.)	165.4	143.1	129.6	121.4	**117.0**	115.5	116.5

Table 7: χ^2 against the exponent α in $g^\dagger \propto E(z)^\alpha$ ($p = 0.5$). Allen: best $\alpha = 1.50$ (edge of the grid), 1σ [1.35, 1.50], 2σ [1.10, 1.50]. Mantz: best 1.25, 1σ [1.05, 1.50], 2σ [0.85, 1.50]. The absolute χ^2 per degree of freedom (1.7 and 3) shows that the errors do not account for the scatter; only the differences carry information. The two samples share several clusters, so the χ^2 are not additive.

5.2 Galaxies at $z \simeq 1$–2

Übler et al. (2017, 135 KMOS^{3D} galaxies) give median properties in three bins (Table 8). We model each bin as an exponential disk with the median baryonic mass and effective radius ($R_d = R_e/1.678$, all baryons in the same disk), compute $v_{\rm circ}(r) = \sqrt{K_{\rm eff}\, g_{\rm bar}\, r}$ with $n = 1/2$, $s = 1$ and take its maximum, a quantity comparable in definition to the modelled $v_{\rm circ,max}$ of those authors, which includes a pressure-support correction.

Bin	$\log M_{\rm bar}$	R_e	$v_{\rm obs}$	$v_{\rm RGD}$ (α=1)	$v_{\rm RGD}$ (α=0)	$x_{\rm peak}$
$z \simeq 0.9$	10.62	4.8	239	215 (−10%)	193 (−19%)	0.33
$z \simeq 1.5$	10.97	4.9	263	290 (+10%)	244 (−7%)	0.50
$z \simeq 2.3$	10.89	4.0	260	303 (+16%)	239 (−8%)	0.43

Table 8: KMOS^{3D} medians (Übler et al. 2017, Table 1; R_e in kpc, velocities in km s^{-1}) and RGD predictions for $g^\dagger \propto E^\alpha$. Changing n from 0.50 to 0.55 raises v by 2%; enlarging R_e by 30% lowers it by 2%.

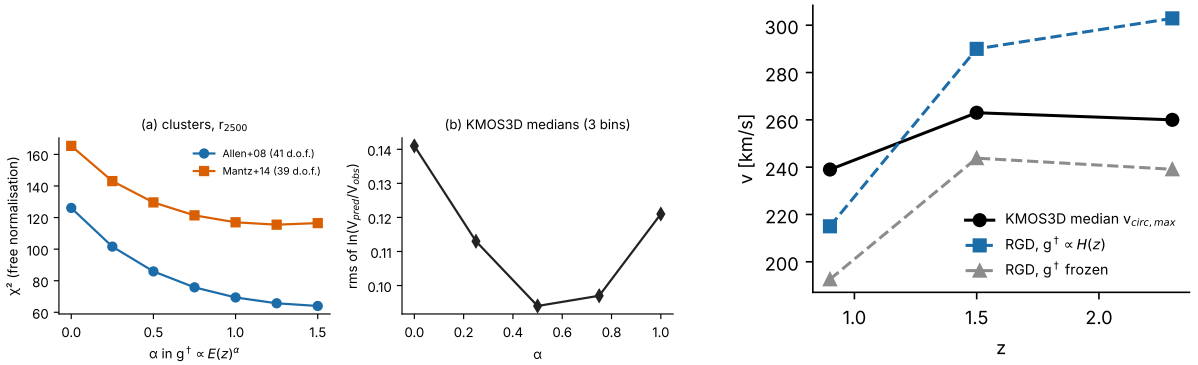


Figure 2: Left: $\chi^2(\alpha)$ for the two cluster samples (a) and rms of $\ln(v_{\rm pred}/v_{\rm obs})$ for the three KMOS^{3D} bins (b). Right: median circular velocities against redshift.

For $g^\dagger \propto H(z)$ the zero-point shift relative to $z = 0$ is $-3.75\Delta \log v = -0.18$, -0.28, -0.38 dex in $\log M$ at fixed velocity for the three bins, a monotonic trend, while the observed offsets relative to the local baryonic Tully–Fisher relation are -0.44 (at $z \simeq 0.9$) and -0.27 (at $z \simeq 2.3$), non-monotonic. The rms of $\ln(v_{\rm pred}/v_{\rm obs})$ is 0.141 for $\alpha = 0$, 0.113 (0.25), 0.094 (0.50), 0.097 (0.75) and 0.121 for $\alpha = 1$. The galaxies therefore favour $\alpha \simeq 0.5$–0.75 and the clusters $\alpha \gtrsim 1$. The two are reconciled only if the high-redshift galaxy systematics (gas masses from scaling relations with 0.2 dex error and no atomic gas, pressure-support corrections of 0.1–0.2 dex, selection that changes the stellar mass across bins) are of order 10% or more, which they are; with three bins we cannot decide whether α depends on system scale. The dark matter fractions inferred in $z \simeq 2$ galaxies are lower than at $z = 0$ (Übler et al. 2017; Genzel et al. 2017), whereas a larger $g^\dagger$ at the same baryonic acceleration produces a larger K; this is the same tension seen from another side and has not been tested quantitatively here.

6 Connecting galaxies to clusters: the ξ switch

The radial term cannot be applied to galaxies: in SPARC an exploratory $s = (\Delta/\Delta_{\rm out})^p$, with Δ taken from the rotation curve itself, degrades the rms from 0.148 to 0.158, 0.235 and 0.468 dex for $p = 0.25$, 0.5 and 1, and the global fit favours $p \simeq 0.09$. A switch based on the local density does not separate the regimes, because inner cluster radii reach galactic densities; the discriminator must be a property of the system. The compactness ξ is such a property: SPARC haloes have $\xi \simeq 0.004$–0.094 (under two halo mass assumptions, A: $M_{500} = M_b/0.10$; B: a stellar-to-halo relation, so ξ for galaxies is imputed, not measured), the 43 groups $\xi = 0.136$–0.29, X-COP 0.34–0.54 and the Allen/Mantz clusters 0.40–0.85. No object lies in $0.094 < \xi < 0.136$. A saturating switch such as $1 - e^{-\xi/0.052}$, the form inherited from $\alpha(\xi)$ in Paper I, turns the term on partially at galactic ξ and was found incompatible with the SPARC residuals.

Must the switch be a sharp step? We fitted $\Theta = [1 + (\xi_c/\xi)^m]^{-1}$ with free (ξ_c, m, p) jointly to the X-COP points, the SPARC galaxies and the 43 groups (Kravtsov stars, $\varphi = 0.7$), weighting the three samples by their number of independent units (13 clusters, 141 galaxies, 43 groups; weighting by points would let the 2803 SPARC points dominate the likelihood). Table 9 gives the best fit for fixed m.

m	$\Delta(-2\ln L)$ A	$\Delta(-2\ln L)$ B	ξ_c (A / B)	p (A / B)
1	37.0	49.6	0.49 / 0.58	0.95 / 1.00
2	9.3	18.6	0.13 / 0.15	0.50 / 0.50
3	1.8	4.4	0.11 / 0.14	0.45 / 0.50
4	**0.0**	**0.0**	0.10 / 0.115	0.45 / 0.45
6	3.3	1.1	0.09 / 0.11	0.45 / 0.45
8	4.5	3.9	0.15 / 0.10	0.45 / 0.45
16	3.8	4.3	0.16 / 0.16	0.45 / 0.45
32 (sharp)	4.6	5.2	0.16 / 0.16	0.45 / 0.45
∞ (200)	6.9	7.4	0.16 / 0.16	0.45 / 0.45
power law $p_0(\xi/0.4)^q$	46.2	53.3	—	$p_0 = 0.40$, $q = 0.85$ / 0.95

Table 9: Joint fit of the switch to X-COP, SPARC and the groups, for halo mass assumptions A and B for the galaxies. $\Delta(-2\ln L)$ is relative to the best m ($-2\ln L = -885.8$ for A). At $m \geq 4$ the rms of the clusters is 0.016, SPARC 0.138–0.142 and the groups 0.074–0.077 dex. The power law has two parameters, as does a step with fixed m.

The data exclude a very smooth switch ($m = 1, 2$) and prefer $m \simeq 4$, but a sharp step costs only 4.6–5.2 in $-2\ln L$, which after charging the extra parameter m is ΔAIC $\simeq +2.6$ to $+3.2$ in favour of the logistic: the sharpness is not determined. The location is likewise loose: $\xi_c = 0.10$–0.12 for $m = 4$, and anywhere in 0.09–0.16 for sharper steps (the likelihood is flat because only a few groups lie at $\xi < 0.16$); the bracket of the first draft, $0.09 \lesssim \xi_c \lesssim 0.14$, rested on the same absence of systems between galaxies and groups and was too narrow. A smooth power law $p(\xi) = p_0(\xi/0.4)^q$ without threshold, with as many parameters as a step, is disfavoured by $\Delta(-2\ln L) = 42$–48 with respect to the step and 46–53 with respect to $m = 4$: the galaxies need $p \simeq 0$ and the clusters 0.45, and the groups lie between, which a power law cannot reproduce without degrading the clusters (rms 0.024–0.026 against 0.016). The switch therefore behaves as a threshold in ξ, of unknown width.

Dependence on the stellar mass of groups. The group amplitude is set by the unmeasured stellar mass inside r_{2500}. With Kravtsov stars ($\varphi = 0.7$) the groups give $p = 0.45$ [0.39, 0.50], consistent with the clusters; with a stellar mass of 9% of the gas (first draft) they give $p = 0.80$

[0.76, 0.85] and the joint fit then degrades the clusters from 0.016 to 0.10 dex. The consistency of p across regimes therefore holds for the Kravtsov relation and fails for the lower stellar masses, and groups cannot yet arbitrate.

Because $\xi \propto (M_{500}H)^{1/3}$ the switch predicts a redshift-dependent threshold mass whose ratio to the $z = 0$ value, 0.57 at $z = 1$ and 0.34 at $z = 2$, is independent of ξ_c:

$M_{500}^{\rm thr}$ $[M_\odot]$	$z = 0$	$z = 1$	$z = 2$
$\xi_c = 0.09$	4.2×10^{12}	2.4×10^{12}	1.4×10^{12}
$\xi_c = 0.115$	8.8×10^{12}	5.0×10^{12}	3.0×10^{12}
$\xi_c = 0.16$	2.4×10^{13}	1.3×10^{13}	8.0×10^{12}

For $m = 4$ the switch goes from 10% to 90% over a factor $\simeq 3$ in ξ, i.e. about 27 in M_{500}, so the thresholds are indicative; for the baryonic Tully–Fisher relation the corresponding zero-point shift is $\Delta \log v = 0.25 \log_{10} E(z)$ (+0.061 dex at $z = 1$, +0.118 at $z = 2$).

7 Comparison with ΛCDM

Table 10 compares the kernel with NFW haloes (Navarro et al. 1997) added to the same baryons, using either the concentration–mass relation of Duffy et al. (2008) or a free concentration.

Regime		Model	free par.	rms [dex]	AIC	note
SPARC, gal. (2736 pts)	141	RGD kernel, $s = 1$	0	0.142	−10670	
		NFW + Duffy $c(M)$	141	0.087	−13066	
		NFW, c free	282	0.059	−14877	
SPARC, gal.	175	RGD kernel	0	0.148	—	2803 pts
		RAR, $a_0 = 1.2 \times 10^{-10}$	0	0.144	—	
		Newton	0	0.518	—	
Groups (43)		NFW + Duffy, norm. to M_{500}	0	0.089	—	mean −0.020
		RGD, $p = 0.45$, Kravtsov stars	1	0.079	—	mean 0.000
		RGD, $p = 0.5$, Kravtsov stars	0	0.081	—	mean +0.019
		RGD, $p = 0.5$, stars 9% of gas	0	0.131	—	mean −0.118
		RGD, $p = 0.8$, stars 9% of gas	1	0.056	—	mean −0.001
X-COP, 39 pts		RGD $p = 0.5$ / fitted	0 / 1	0.021 / 0.016	−301 / −321	Table 3
		NFW + Duffy	13	0.039	−227	

Table 10: Kernel against ΛCDM haloes. Group numbers predict $M(< r_{2500})$ from M_{500}; the ΛCDM line has no free parameter. Kravtsov stars: $\varphi = 0.7$ of the relation inside r_{2500} (best $p = 0.45$ [0.39, 0.50]; $\varphi = 0.5$–1 gives 0.45–0.54); the 9%-of-gas rows are the sensitivity case. φ is not measured. In the X-COP rows the observed baryon fraction is the scaling relation of Paper I, not a measured profile.

ΛCDM fits SPARC rotation curves better (0.087 or 0.059 against 0.142 dex), at the cost of one or two free parameters per galaxy; the kernel has none, and equals the RAR in quality (0.148 against 0.144). In groups the outcome depends on the stellar mass, which is not measured: with the Kravtsov relation the kernel (0.079–0.081 dex) is not distinguishable from ΛCDM with a $c(M)$ prior and no free parameter (0.089), whereas a stellar mass of 9% of the gas gives 0.131 at $p = 0.5$ and needs $p = 0.8$ (0.056). The groups therefore do not discriminate. In X-COP the radial term wins by AIC with a margin of 74 to 94 over NFW + Duffy when only the parameters added to the fixed kernel are counted. Charging also p, ξ_c, m and A (four parameters) the margin is 66 with $N = 39$ and 34 with one independent unit per cluster ($N_{\rm eff} = 13$, same charge for NFW). Neither count is neutral: the 13 NFW parameters are per-cluster masses, whereas the

kernel predicts masses from baryons but inherits A from 59 systems that include these clusters; and the comparison is within the calibration sample and with the Paper I baryon fraction. A positive offset of +0.15 dex appears in SPARC galaxies with $V_{\rm max} < 70$ km s^{-1} (312 points, rms 0.25), which we find not to depend on $M_{\rm bar}$ and interpret as a selection effect of grouping by the observed maximum velocity. We therefore describe the kernel as competitive in clusters and parsimonious in galaxies, not as superior in fit.

8 Tensions and tests that did not close

(1) Baryon fraction against mass at r_{2500}. The kernel predicts $f_{\rm bar}(r_{2500})$ rising with M_{2500} by +0.35 per dex, against a measured mass slope of the observed fraction of -0.04 ± 0.09 (Allen) and -0.09 ± 0.07 (Mantz). In the Vikhlinin sample ln(obs/pred) has a trend of −0.31 per dex of M_{2500} with $p = 0.5$ (median obs/pred 0.84, rms$_{\rm ln}$ 0.225 over twelve clusters) and −0.35 with $p = 0.46$ (0.76, 0.29). The effect worsens with realistic stellar masses and is independent of the form of $n(\xi)$.

(2) Absolute normalisation. At r_{500}, where p plays no role, obs/pred is 0.95 for Vikhlinin (Chandra) and 1.17 for X-COP (XMM-Newton), a $\sim$ 20% difference that is larger than the radial effect under test; at r_{200} the X-COP value is 1.24. Known cross-calibration differences between the two instruments are of this order and were not modelled. In terms of the ratio tests of Sect. 4.3 the two samples give -0.10 ± 0.06 and $+0.03 \pm 0.04$ for $p = 0.5$; their difference of 0.13 in ln is of the size of the zero-point offset and sets the scale of the cross-sample systematic.

(3) A mass-dependent exponent. The form $p(\xi) = p_0(\xi/0.4)^q$ with $q > 0$ (bootstrap $q = 0.37$ [0.31, 0.43], $p_0 = 0.46$ over Allen and Mantz) removes the trend of tension (1) ($-0.31 \to -0.06$, Fig. 3), but predicts a mass trend of $f_{\rm bar}(r_{200})/f_{\rm bar}(r_{500})$ of +0.11 against −0.12 observed and of the within-cluster r_{2500}/r_{500} ratio of −0.25 against −0.07. It is not supported and is discarded as a hypothesis; the simplest reading remains constant $p \simeq 0.5$.

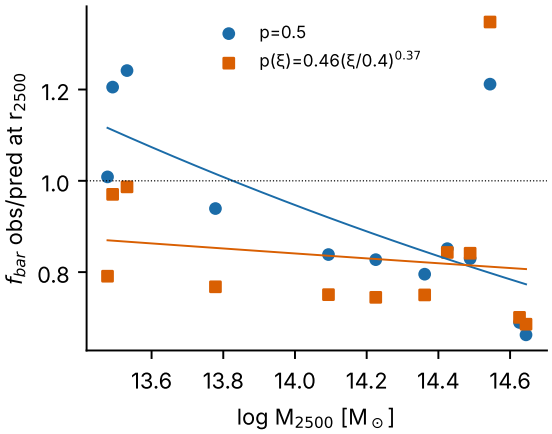


Figure 3: Vikhlinin et al. clusters: obs/pred baryon fraction at r_{2500} against $\log M_{2500}$ for constant $p = 0.5$ (trend −0.31 per dex) and for $p(\xi) \propto \xi^{0.37}$ (trend −0.06). Lines are log-linear fits.

(4) Hydrostatic equilibrium and lensing. All cluster masses but CLASH are hydrostatic. Hydrostatic masses measure $g_{\rm obs}$ without bias if the gas is in equilibrium in the potential of Eq. (1), but non-thermal pressure (5.9% at r_{500} and 10.5% at r_{200} in X-COP) and departures from equilibrium were not modelled and the temperature and density profiles predicted by the

kernel were not computed. A mass-dependent bias of the sign needed to close tension (1) would require hydrostatic masses to overestimate the true mass at high mass, opposite to the usual expectation from non-thermal pressure. Weighing-the-Giants calibration finds $M_L/M_X = 0.96\,(+0.06/-0.09)\pm 9\%$ (syst.) at r_{2500}, with a mass index $\beta = 0.78\,(+0.17/-0.14)$ (assuming $M_L \propto M_X^{\beta}$), less than 2σ from 1 (Mantz et al. 2016); closing the trend would need $\beta \simeq 0.87$. For the 12 clusters common to CLASH and Allen, $M_L/M_X(r_{2500})$ has median 1.21 (weighted mean 1.09, scatter 0.35 in ln), obs/pred is 0.93 with lensing masses and 1.10 with X-ray masses ($\mathrm{rms}_{\ln}$ 0.24 and 0.34). The lensing sample is small, high-mass and noisy, the gas was rescaled to the lensing aperture assuming $M_{\rm gas} \propto r^{1.2}$, and noise in M_L enters both axes, so the mass trend itself is not informative there. Lensing calibration therefore reduces but does not decide the matter.

(4b) Sensitivity to a hydrostatic bias. We set $M_{\rm HSE} = (1-b)M_{\rm true}$ and applied the kernel to $M_{\rm true}$. (a) X-COP, 39 points, $b(w) = b_{500}w^{\kappa}$ with $\kappa = 0$ or 1 and $-0.1 \le b_{500} \le 0.3$: the best-fit p stays within 0.40–0.49; the rms is 0.016–0.028 dex for the new term, 0.037–0.057 for v4 and 0.110–0.154 without radial term, so the ordering of the three models does not depend on the bias. (b) With measured gas fractions ($f_{\rm bar} = M_{\rm gas}/M_{\rm true}$) the best-fit p rises with b: Vikhlinin (ratio test) 0.53 for $b = 0$, 0.56 for $b_{2500} = 0.10$–0.15, 0.60 for 0.20–0.25 and 0.63 for 0.30; Allen 0.48, 0.53, 0.57 and 0.60 for $b = -0.1, 0, 0.1, 0.2$. At fixed $p = 0.5$ the Vikhlinin ratio test moves from -0.10 ± 0.06 to -0.21 ± 0.07 ($b_{2500} = 0.15$) and -0.33 ± 0.06 (0.20), while the v4 law degrades from -0.61 to -1.2. A bias of 0.17, the value implied by the CLASH–Allen median $M_L/M_X = 1.21$, thus corresponds to $p \simeq 0.57$, and the Mantz et al. calibration ($M_L/M_X = 0.96$, $b \simeq -0.04$) to $p \simeq 0.52$: both lie inside the sample-to-sample dispersion of Sect. 4.3. (c) A positive bias does not remove the mass trend of tension (1): fitting $b = b_0 + b_1(\log M_{2500} - 14.4)$ to Allen requires $b_1 \simeq -0.2$ to -0.3 per dex, i.e. $b(10^{15}\,M_\odot) \simeq -0.12$ to -0.21, hydrostatic masses overestimating the true mass in the most massive clusters. The conclusion on the necessity of the radial term and its ordering against v4 is robust to hydrostatic bias; the amplitude p and the mass trend are not.

(5) High-redshift galaxies. Sect. 5.

9 Extension to cosmological scales

9.1 The kernel applied to linear perturbations

For a top-hat baryonic perturbation δ_b of physical radius R, $g_{\rm bar} = \frac{1}{2}\Omega_b(a)H^2\delta_b R$ and

$$x = \frac{g_{\rm bar}}{g^\dagger} = \pi\,\Omega_b(a)\,\delta_b\,\frac{R}{c/H}. \tag{4}$$

Over the history of the Universe and for $R = 0.1$–100 Mpc comoving, $x = 10^{-8}$–4×10^{-3}, always deep in the low-acceleration regime. At $z = 1100$ with $\delta_b = 10^{-4}$, $K = 7282$, 2303, 729 and 231 for $R = 0.1, 1, 10, 100$ Mpc, whereas mimicking cold dark matter needs $K = \Omega_m/\Omega_b = 6.1$. A toy growth equation $\delta'' + (2 + d\ln H/d\ln a)\delta' = \frac{3}{2}\Omega_b K(x(\delta))\delta$ from $z = 1100$ gives $\delta(z=0) = 123$, 127 and 134 for $\delta_i = 10^{-5}$, 10^{-4}, 10^{-3} at $R = 10$ Mpc comoving (it scales as $1/R$: 1.2×10^3 at 1 Mpc, 13–15 at 100 Mpc), against 0.0076, 0.076 and 0.76 in ΛCDM. The result is nearly independent of the initial amplitude (the force scales as $\delta^{1/2}$), so the primordial amplitude A_s would no longer control the spectrum and the fluctuations would not be linear and Gaussian. The toy neglects photon pressure and baryon–photon coupling before decoupling, which would delay growth, but the excess is 10^3–10^5.

9.2 Dilemma and damper

If the kernel acts on cosmological perturbations it overshoots; if it acts only on bound systems, baryons alone grow by a factor 13 between $z = 1100$ and today against 760 in ΛCDM ($\delta = 0.013$ instead of 0.76 for $\delta_i = 10^{-3}$) and no galaxy forms, and the CMB acoustic peaks cannot be produced without a potential well that does not couple to photons. A damper based on the baryon density contrast,

$$K_{\rm eff} = K_c\,(1 - D) + K(x)\,D, \qquad D = \mathrm{expit}\left[m \ln \frac{\rho_b/\rho_c}{\Delta_v}\right], \qquad K_c = \frac{\Omega_m}{\Omega_b} \simeq 6.12, \tag{5}$$

with $\Delta_v = 30$ in ρ_b/ρ_c (total $\Delta \simeq 200$) and $m = 4$ leaves bound systems unchanged and gives $\delta(z = 0) = 0.7604$ for $R = 1$, 10 and 100 Mpc, identical to ΛCDM within 10^{-7} in the growth rate. This equality holds by construction, since $\rho_b/\rho_c \lesssim 0.1$ in the linear regime and $D \to 0$: it is consistent but not a test, and K_c is a postulate. A hard cap $K \leq K_c$ everywhere is excluded, since at r_{500} the kernel already gives $K \simeq 7.5 > 6.12$ ($f_{\rm bar} = 0.133$ against the cosmic 0.163); a cap would set $f_{\rm bar}(r_{500}) = 0.163$ against the observed ~ 0.13. That clusters sit below the cosmic baryon fraction without feedback is a qualitative consequence of $K > K_c$ in bound systems; we have not tested it against a specific measurement of the baryon depletion.

9.3 CMB

Reproducing the acoustic peaks requires a perturbation theory with baryon–photon coupling and a relativistic completion; kernel-type enhancements of the baryon force do not reproduce the third peak and the odd–even peak ratio, and the theories that do (e.g. Skordis & Złośnik 2021) add a field that behaves as pressureless matter. We make no claim about the CMB or large-scale structure.

10 Discussion

10.1 Evolution with respect to Paper I

What improves. (i) The radial law $n(\xi)w^{-\alpha(\xi)}$, with two parameters calibrated on the same X-COP data and an exponent that exceeds 1 inside r_{500}, is replaced by a scale shift with one amplitude that stays finite there (X-COP rms 0.037 → 0.016–0.021 dex, in-sample). (ii) With p frozen, the new term outperforms the v4 law inside r_{500} (Vikhlinin, Allen and the groups, where v4 loses solutions or diverges) and is equivalent to it between r_{500} and r_{200} (Sect. 4.3). (iii) The program now carries an audit: provenance of each parameter, bootstrap intervals, sensitivity to hydrostatic bias, a hierarchical model of the amplitude (intrinsic scatter 0.05 among clusters) and an independent confirmation of the threshold in ξ (Sects. 4.3–4.4). (iv) The limits, the tensions and the cosmological failure of the linear kernel are stated rather than left implicit.

What is weaker or more modest. (i) The headline numbers of Paper I (0.005 dex in-sample, 0.013 dex with A calibrated without CLASH, and a residual correlation $r = 0.999$ that Paper I itself decomposed into an algebraic part and a 0.912 → 0.999 gain attributed to evolution) are not used here as evidence. (ii) The model has more parameters: the amplitude p and a switch in ξ (ξ_c, m) that Paper I did not need, because it did not separate galaxies from groups. (iii) The evidence for $g^\dagger \propto H(z)$ is weaker than in Paper I: the best exponent in clusters lies at the edge of the grid and galaxies at $z \simeq 1$–2 prefer a weaker evolution (Sect. 5). (iv) In groups the outcome depends on an unmeasured stellar mass, so they do not discriminate between models.

What remains open. No blind test has been made; the mass trend of the baryon fraction at r_{2500} persists; the redshift dependence of p cannot be separated from that of $g^\dagger$ with the present data; there is no covariant formulation; and the cosmological extension reproduces ΛCDM growth only by postulate.

Assessment. The gain in observational performance over Paper I is modest and concentrated in the radial sector of clusters; the gain in methodological rigour is large. Paper I showed a kernel that fitted well; this version shows what the kernel predicts, where it fails and with what uncertainty, which turns the program from a claim of fit into a testable hypothesis. A blind test on new samples with the archived frozen prescription is the next step needed before the claims can be stated more strongly.

10.2 What the program claims

1. In bound systems with symmetric equilibrium the kernel, with λ, A, $g^\dagger$ fixed on galaxies and groups, describes the baryon fraction across four orders of magnitude in mass with one radial amplitude ($p_0 = 0.53$, with intrinsic scatter 0.05 among clusters and 0.17 among groups, Sect. 4.4) and one switch in ξ ($\xi_c \simeq 0.09$–0.16, width $m \geq 3$).
2. The radial term $[\Delta/\Delta_{500}]^p$ replaces the two-parameter exponent modulation, reduces the X-COP rms from 0.037 to 0.016–0.021 dex (in-sample) and, unlike the former, extends inside r_{500} and, with p frozen, to r_{2500} and r_{200} on held-out (not blind) data, with a predicted baryon fraction good to $\sim 13\%$ (intrinsic scatter) in clusters and much less in groups.
3. $g^\dagger \propto H(z)$ is the preferred evolution in clusters and is not excluded in galaxies at $z \simeq 1$–2 within current systematics.

This is a phenomenological statement of restricted scope, defensible from the data used and not derived from a Lagrangian.

10.3 What the program does not claim

It does not claim that dark matter does not exist, nor that it replaces dark matter in cosmology: the CMB, linear growth and primordial perturbations are untouched, and the damper of Eq. (5) reproduces ΛCDM growth by postulate. It does not claim superiority in fit over ΛCDM haloes, which fit SPARC better and, depending on the unmeasured stellar mass, may fit the groups better. The tensions of Sect. 8 are open.

10.4 Comparison with competing approaches

Against MOND (Milgrom 1983) and the RAR (McGaugh et al. 2016) the transition scale evolves as $cH(z)/2\pi$ and the exponent depends on the compactness of the equilibrium; against ΛCDM the kernel is a baryon-only prescription with no per-system halo parameters in the tested regimes, but without cosmology. Observationally the two differ by the redshift evolution of the transition acceleration and by the dependence of the baryon fraction on ξ and radius. Covariant relativistic MOND completions (e.g. AeST) address the CMB by adding a field; the kernel does not.

10.5 Agenda for falsification

1. **Radial parameter.** Resolved gas and lensing profiles (X-COP, eROSITA, XXL) should give $p = 0.5 \pm 0.1$ independent of ξ and mass; given the present intrinsic scatter of 0.05

among clusters (0.17 among groups), a single system or sample outside 0.5 ± 0.15 is not decisive, but $|p-0.5| > 0.15$ in several independent samples or a significant mass dependence falsifies the scale shift. Apply the archived frozen prescription before looking at the data.

2. **Evolution of $g^\dagger$.** Clusters at $z > 0.5$ with measured gas fractions (SPT, XMM) and larger samples of $z \simeq 1$–2 rotation curves should converge on a common α; independent samples preferring $\alpha < 0.5$ at 2σ in clusters, or $\alpha > 1$ in galaxies, falsify $g^\dagger \propto H(z)$.

3. **The switch.** Systems with $0.09 < \xi < 0.16$ must show a transition of finite width (for $m = 4$, 10% to 90% over a factor 3 in ξ); galaxies with $\xi > 0.16$ should show $s \neq 1$, and systems with $\xi < 0.09$ should show none. The threshold masses of Sect. 6 give the redshift dependence.

4. **Equilibrium in the kernel potential.** Predict $T(r)$ and $n_{\rm gas}(r)$ for the 12 X-COP and 12 Vikhlinin clusters in the potential of Eq. (1) and compare with the measured profiles, including non-thermal pressure.

5. **Lensing at low mass.** Per-cluster lensing masses (CCCP, LoCuSS, WtG) below $M_{2500} \simeq 2 \times 10^{14}\, M_\odot$ to decide the mass trend of Sect. 8; measured $M_*(< r_{2500})$ in groups to fix φ.

6. **Systems not tested here.** El Gordo ($\xi \simeq 0.73$) is an active merger and violates the equilibrium premise; ATLAS-3D/SLUGGS early-type galaxies and DESI rotation curves remain open from Paper I.

7. **Covariant formulation.** Without an action the program has no predictions for gravitational waves, the CMB or strong fields.

11 Conclusions

The kernel of Paper I, extended with a single-amplitude radial term, describes bound systems from galaxies to clusters better than before; it remains a phenomenological map, not a cosmological theory.

Established.

1. The radial term is necessary: without it the kernel mispredicts the baryon fraction at r_{2500} by a factor 3–4 and at r_{200} by a factor 2.4.

2. $s = [\Delta/\Delta_{500}]^p$ with $p \simeq 0.45$–0.5 describes it with one amplitude: X-COP rms 0.126 $\rightarrow$ 0.016–0.021 dex (v4 law: 0.037; in-sample), and with $p = 0.5$ frozen the ratio tests give $\ln(\text{obs/pred}) = -0.10 \pm 0.06$ at r_{2500} (Vikhlinin; v4: -0.61 ± 0.21) and $+0.03 \pm 0.04$ at r_{200} (Ettori; v4: $+0.02 \pm 0.06$).

3. The v4 law $w^{-\alpha(\xi)}$ is adequate for $w \gtrsim 0.7$ in X-COP but fails inside r_{500} ($n > 1$) and does not meet the v4 criterion of 0.10 dex at r_{2500}.

4. $g^\dagger \propto H(z)$ is preferred by Allen and Mantz gas fractions; a frozen $g^\dagger$ costs $\Delta\chi^2 = 57$ and 48.

5. A threshold in ξ ($\xi_c = 0.09$–0.16, width $m \geq 3$) connects galaxies, groups and clusters; a smooth power law $p(\xi)$ is disfavoured by $\Delta(-2\ln L) \simeq 50$, and the threshold mass scales with redshift by 0.57 ($z = 1$) and 0.34 ($z = 2$).

Not established.

1. The amplitude p is not an exact constant: the hierarchical model gives an intrinsic scatter of 0.05 among clusters ($\sim$ 13% in the predicted baryon fraction) and 0.17 among groups, with sample-to-sample best-fit values of 0.45–0.60; no dependence on mass, redshift or instrument survives individually once the ξ switch is included, but the redshift coefficient is degenerate with the evolution of $g^\dagger$.
2. The sharpness and location of the galaxy–group switch are loosely constrained, and the group amplitude depends on an unmeasured stellar mass.
3. The baryon fraction at r_{2500} rises with mass more than observed (-0.31 per dex in obs/pred); lensing calibration shifts it by $\sim 1\sigma$.
4. The absolute normalisation differs by $\sim 20\%$ between the Chandra and XMM samples (0.95 against 1.17 at r_{500}).
5. A mass-dependent $p(\xi) \propto \xi^{0.37}$ closes the former but predicts the wrong sign of the $r_{500} \to r_{200}$ trend, and is discarded.
6. Median KMOS^{3D} galaxies prefer $g^\dagger \propto E^{0.5\text{–}0.75}$ against $\alpha \gtrsim 1$ in clusters, within the $\sim 10\%$ systematics of the former.
7. Applied to linear perturbations the kernel boosts gravity 10^2–10^3 times; the damper recovers ΛCDM growth only through the postulated plateau $K_c = \Omega_m/\Omega_b$; no CMB statement is made.

Position relative to ΛCDM. In clusters the kernel is competitive (X-COP AIC -301 to -321 against -227, in-sample; margin 34–66 after charging p, ξ_c, m, A); in galaxies it is parsimonious (zero parameters) but fits worse than haloes (0.142 against 0.087 or 0.059 dex); in groups the kernel and ΛCDM are indistinguishable with Kravtsov stellar masses and ΛCDM is better with lower ones; in cosmology the kernel makes no competitive claim. The defensible claim is an empirical map of bound systems that any covariant theory must reproduce in the weak-field stationary limit.

Practical value of parsimony. The model has no per-system free parameter: given the baryonic profile and the equilibrium geometry of a system, $K_{\rm eff}$ returns $g_{\rm obs}$ (or the baryon fraction) from five global numbers, λ and $g^\dagger$ fixed on galaxies and A, p and the switch fixed on groups and clusters, all frozen in the archived prescription. Three practical consequences follow, within the accuracy documented above. (i) *Forecasting without a halo fit.* A rotation curve, a baryon fraction at r_{2500} or the threshold mass at which a system leaves the galactic regime can be stated before the data are taken, with an error budget (0.14 dex in galaxies, $\sim 13\%$ intrinsic in the cluster baryon fraction), whereas an NFW description needs one or two free parameters per galaxy (141–282 for the SPARC sample) or one mass per cluster, and can absorb a mismatch that the kernel cannot. (ii) *Cheap and auditable computation.* Each prediction is a single root-finding with no sampling, so it can be applied to large surveys or to sparse and high-redshift data where halo parameters are unconstrained, and any discrepancy is attributable to one of a handful of numbers rather than to a degenerate fit. (iii) *Falsifiability.* The same scarcity of parameters is what exposed the tensions of Sect. 8 and the dispersion of p in Sect. 4.3; each further sample either confirms the frozen values or breaks them. Parsimony is a virtue only while the accuracy holds: the kernel fits galaxies worse than haloes (0.142 against 0.087 dex), the amplitude p has an intrinsic scatter ($\sim$ 13% in the predicted baryon fraction for clusters, larger for groups), and it needs a measured baryonic profile as input. Its practical role is therefore a first-order, reproducible estimator of dynamical masses in bound systems, and a sharp test for competing descriptions, not a replacement of halo fitting where precision is required.

Data and code availability

All scripts, the transcribed tables and the figure generator are in the public repository `https://github.com/Maia-KarlosMarden/rgd-kernel` (folder `v5`). SPARC, X-COP, CLASH, Allen, Mantz, Vikhlinin, Lovisari, Sun and KMOS^{3D} data are public from the original publications; the transcribed tables should be checked against the originals before reuse.

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A Reproducibility notes

- Scripts are run from inside the `v5` folder (several use `exec(open(...))`); `v4_vs_v5.py` reproduces Table 3, `v4_indep.py` the independent radial tests, `alpha_scan.py` Table 7, `highz_tf.py` Table 8, `lcdm_compare.py` Table 10, `cmb_regime.py` and `damper_test.py` Sect. 9.
- Referee-type checks: `ref31.py` (bootstrap, frozen-p tables, best-fit p per sample; Sect. 4.3), `ref32.py` (hydrostatic bias; Sect. 8), `ref34.py` (switch width and location; Table 9), `ref41.py`, `ref41b.py`, `ref41c.py`, `ref42.py` (hierarchical model of p and its correlation with z; Sect. 4.4). `predict_frozen.py` implements the frozen prescription ($p = 0.5$, $\xi_c = 0.115$, $m = 4$, $\varphi = 0.7$, λ, A, $g^\dagger = cH/2\pi$) for new samples.
- SPARC Table 2 is not redistributed; set the environment variable `SPARC_TABLE2`.
- Mantz fractions are shell values and Allen and Vikhlinin cumulative; Mantz M_{2500} was validated against the critical-density definition, Vikhlinin M_{500} likewise. Cosmology: flat, $H_0 = 70$, $\Omega_m = 0.3$ throughout except where stated (Vikhlinin: $h = 0.71$; CLASH: $\Omega_m = 0.27$; the differences were not propagated).
- The X-COP rows labelled "scaling" follow Paper I's $f_{\rm bar} = 1.09 f_{\rm gas}(M_{500})$ with $f_{\rm gas} = 0.0708(M_{500}/10^{13}M_\odot)^{0.135}$; the measured-profile test is the Ettori et al. Table 2 analysis.

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